

PANJAB UNIVERSITY, CHANDIGARH-160014 (INDIA)

(Estd. under the Panjab University Act VII of 1947—enacted by the Govt. of India)

FACULTY OF ARTS

SYLLABI

FOR

M.A. ECONOMICS (SEMESTER SYSTEM)
W.e.f. SESSION 2019-20

(1st & 2nd Semesters)

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APPLICABILITY OF REGULATIONS FOR THE TIME BEING IN FORCE

Notwithstanding the integrated nature of a course spread over more than one academic year, the regulations in force at the time a student joins a course shall hold good only for the examinations held during or at the end of the academic year. Nothing in these regulations shall be deemed to debar the University from amending the regulations subsequently and the amended regulations, if any, shall apply to all the students whether old or new.

GUIDELINES FOR CONTINUOUS INTERNAL ASSESSMENT (20%) FOR REGULAR STUDENTS
OF POST-GRADUATE COURSES of M.A. Economics (Semester System)
(Effective from the First Year Admissions for the Academic Session 2005-2006)

1. The Syndicate has approved the following guidelines, mode of testing and evaluation including Continuous Internal Assessment of students :
   (i) Terminal Evaluation : 80 %
   Continuous
   (ii) Assessment : 20 %
   (iii) Continuous Assessment may include written assignment, snap tests, participation in discussions in the class, term papers, attendance etc.
   (iv) In order to incorporate an element of Continuous Internal Assessment of students, the Colleges/Departments will conduct **one** written test as quantified below :

(a)	Written Test	:	25 (reduced to 5)
(b)	Snap Test	:	25 (reduced to 5)
(c)	Participation in Class Discussion	:	15 (reduced to 3)
(d)	Term Paper	:	25 (reduced to 5)
(e)	Attendance	:	10 (reduced to 2)
		Total :	**100 reduced to 20**

2. Weightage of 2 marks for attendance component out of 20 marks for Continuous Assessment shall be available only to those students who attend 75% and more of classroom lectures/seminars/workshops. The break–up of marks for **attendance component** for theory papers shall be as under :

Attendance Component			*Mark/s for Theory Papers*
(a)	75 % and above upto 85 %	:	1
(b)	Above 85 %	:	2

3. It shall **not be compulsory** to pass in Continuous Internal Assessment. Thus, whatever marks are secured by a student out of 20% marks, will be carried forward and added to his/her score out of 20 %, i.e. the remaining marks allocated to the particular subject and, thus, he/she shall have to secure pass marks both in the University examinations as well as total of Internal Continuous Assessment and University examinations.

4. Continuous Internal Assessment awards from the affiliated Colleges/Departments must be sent to the Controller of Examinations, by name, **two weeks before** the commencement of the particular examination on the Performa obtainable from the Examination Branch.

SPECIAL NOTE:

(i) The theory paper will be of 80 marks and 20 marks will be for internal assessment.

(ii) In the case of Postgraduate Courses in the Faculties of Arts, Science, Languages, Education, Design & Fine Arts, and Business Management & Commerce, falling under the purview of Academic Council, where such a provision of Internal Assessment/Continuous Assessment already exists, the same will continue as before.

OUTLINES OF TESTS, SYLLABI AND COURSES OF READING FOR M.A. ECONOMICS (SEMESTER SYSTEM) FOR THE EXAMINATION OF 2019-2020

Course Structure:

The M.A. in Economics will be a four semester full time programme. The course will be organized over two years, each year consisting of two semesters—the summer semester and the winter semester.

The programme of study consists of sixteen courses out of which **12** are core courses—**compulsory** for all students—and **4** are **elective courses**. Each of these courses will carry 100 marks, of which 20 marks will be for internal assessment and 80 marks for the end semester examinations to be held in December (Semesters I and III) and May (Semesters II and IV). The teaching hours will be 50 per course per semester. The courses will be offered according to the following schedule:

SEMESTER-I

All Compulsory Papers :

MAECO-101	Micro Economics-I	100 Marks
MAECO-102	Macro Economics-I	100 Marks
MAECO-103	Quantitative Methods-I	100 Marks
MAECO-104	International Economics	100 Marks

SEMESTER-II

All Compulsory Papers :

MAECO-201	Micro Economics-II	100 Marks
MAECO-202	Macro Economics-II	100 Marks
MAECO-203	Quantitative Methods-II	100 Marks
MAECO-204	Public Finance	100 Marks

SEMESTER–I

PAPER-MAECO-101: MICRO ECONOMICS-I

Max. Marks	: 100
Theory	: 80 Marks
Internal Assessment	: 20 Marks
Time	: 3 Hours
Teaching Hours	: 50

Objective:

The objective of the paper is to rigorously and comprehensively equip the students with theoretical concepts, methodology and process of reasoning involved in analyzing economic behavior of individuals, firms and markets using, in general, a static and partial equilibrium framework.

The students would be evaluated at the end of each semester through subjective type questions/ answers (both short and essay type). The scripts would be evaluated by the examiners having adequate postgraduate teaching experience in the paper/option concerned.

Pedagogy of the Course Work:

The course relies on a combination of lectures, solving problems, and discussing of academic articles or real-life situations. Teacher will assign topic for assignments on contemporary themes and issues from the syllabi. Special tutorials/contact hour for one-to-one student-teacher interactions.

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES:

1. The syllabus of this paper has been divided into four units.
2. There shall be 9 questions in all.
3. The first question, would be **compulsory**, shall be short answer type (word limit 25-30 each). It would carry 15 short questions, spread over the entire syllabus. The candidate will be required to attempt any 10 short answer type questions. Each short answer type question would carry 2 marks ($10 \times 2 = 20$).
4. Rest of the paper shall contain 4 units. Each unit shall have two questions and the candidates shall be required to attempt one question from each Unit— 4 in all. Each question shall carry **15** marks ($15 \times 4 = 60$).

UNIT-I

Central Ideas of Economics:

How do People Make Decisions? People: face 'Trade Offs'; Incur Opportunity Costs and Gain Benefits of Decisions; Think at the Margin and Respond to Incentives. How do People Interact? Specialization and Exchange can make each party 'Better Off' Market often work efficiently, The Invisible Hand; Three Coordination Tasks in the Economy: Which Commodity be Produced and How much of it? Who should produce it and how? Who should Consume and How much of it?

Methodology of Economics as a Social Science: Alternative Processes of Construction of Positive Economic Theories.

Equilibrium: Concept, Stability, Existence and Uniqueness of Equilibrium.
Types: Static and Dynamic Equilibrium. Partial and General Equilibrium.
Stability Analysis: Marshallian and Walrashallian Approaches.

UNIT- II

Analysis of Consumer Choice under Certainty:

Cardinal Analysis,
Ordinal Utility Approach; Revisiting Consumer Equilibrium and Effects of Changes in Relative Prices and Money Incomes and Derivation of 'Marshallian' and 'Compensated' Demand Curves.

Consumer Surplus: Hicks' and Marshall's Approach.

Application of Indifference Curves: Cash Subsidy Vs. Kind Subsidy: Income Leisure Choice; Inter-temporal Choice of Consumption.
Revealed Preference Theory and Consistency of Choice.

UNIT- III

Market Demand: The Role of Network Externalities; 'Bandwagon', 'Veblen' and 'Snob' Effects.

Analysis of Consumer Choice under Uncertainty:
Measurement of Risk. Expected Utility and Preferences towards Risk. Why do people gamble (insure)? Risk Return Trade Off. Reducing Risk.

Analysis of Consumer Behavior under Asymmetric Information:
Hidden Actions: Principals, Agents and Moral Hazard.
Hidden Characteristics: Adverse Selection Problem. Market and Government Responses to the Problems of Moral Hazard and Adverse Selection.

UNIT-IV

Theory of Production and Costs:
Production, Input Choice and Cost with One Variable Input. Law of Returns to Variable Proportions and Choice of Optimum Quantity of an Input. Derivation of Short Run Cost Curves from Productivity Curves of a Variable Factor of Production. Reserve Capacity and the Shape of Short Run Cost Curves.

Multiple Input Decisions: The Choice of Optimal Techniques of Production and Factor Combinations. Factor Price Changes: Substitution and Output Effects. Elasticity of Input Substitution. Constant, Increasing and Decreasing Returns to Scale. Expansion Path and derivation of Long Run Cost of Production Curves.

Economies and Diseconomies of Scale*:* Shape of Short and Long Run Costs Curves: Theory versus Empirical Evidence. Multi Product Firms and Dynamic Changes in Costs: Economies of Scope; The Learning Curve Analysis.

Recommended Readings:

Essential Readings:

Maddala, G. S. & Miler, E. (2004*). Micro-Economic Theory and Applications*, Tata McGraw Hill.

Mankiw, G. N. (2017). *Principles of Economics (*3rd ed.). Thomson.

Pindyck, Robert, S., Rubinfel, D. L. & Gupta, P.L. (2017). *Micro-Economics.* New Delhi: Prentice Hall of India.

Further Readings:

Boumol, W. J. & Blinder, A. S. (2007*). Macroeconomics: Principles and Policy* (9th ed.). Thomson.

Briet, W. & Hochman, H. M. (1977). *Readings in Micro Economics*. Dryden Press.

Ferguson, C. E. & Gould, J. P. (1989). *Micro Economic Theory* (6th ed.). All India Book Seller.

Koutsoyiannis, A. (2016). *Modern Micro Economics*. London: McMillan Press.

Ryan, W. J. L. & Pearce, D. W. (1989). *Price Theory*. New Delhi: McMillan India.

Salvatore, D. (2007). *Microeconomics: Theory and Applications*, (4th ed.). Oxford University Press.

PAPER- MAECO-102: MACRO ECONOMICS-I

Max. Marks	: 100
Theory	: 80 Marks
Internal Assessment	: 20 Marks
Time	: 3 Hours
Teaching Hours	: 50

Objective:

Macroeconomics or aggregative economics analysis establishes the functional relationship between the large aggregates. The aggregate analysis has assumed such a great significance in recent times that a prior understanding of macroeconomic theoretical structure is considered essential for the proper comprehension of the different issues and policies. Macroeconomics now is not only a scientific method of analysis; but also a body of empirical economic knowledge.

The paper entitled "Macro-Economics-I" equips the students at the postgraduate level to understand systemic facts and latest theoretical developments for empirical analysis.

The students would be evaluated at the end of each semester through subjective type questions/answers (both short and essay type). The scripts would be evaluated by the examiners having adequate postgraduate teaching experience in the paper/option concerned.

Pedagogy of the Course Work:

The course relies on a combination of lectures, solving problems, and discussing of academic articles or real-life situations. Teacher will assign topic for assignments on contemporary themes and issues from the syllabi. Special tutorials/contact hour for one-to-one student-teacher interactions.

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES:

1. The syllabus of this paper has been divided into four units.

2. There shall be 9 questions in all.

3. The first question, which would be **compulsory**, shall be short answer type (word limit 25-30 each). It would carry 15 short questions, spread over the entire syllabus. The candidate will be required to attempt any 10 short answer type questions. Each short answer type question would carry 2 marks ($10 \times 2 = 20$).

4. Rest of the paper shall contain 4 units. Each unit shall have two questions and the candidates shall be required to attempt one question from each Unit— 4 in all. Each question shall carry **15** marks (15 × 4 = 60).

UNIT-I

(Fundamental Theories of Income and Employment Determination)

***Income and Employment Determination*:** Integrated Classical and Keynesian Models of Income and Employment Determination; commodity, money (including bond market of Keynes), and labour markets.
Wage-Price Flexibility and Automatic Full Employment: Classical Versus Keynesian Approach.

UNIT-II

(Consumption Theories)

Consumption and Consumption Function: Keynes Consumption and saving functions under Psychological law of consumption, Consumption Puzzle: Absolute Income hypothesis, Relative Income hypothesis, Permanent Income hypothesis and Life Cycle Hypothesis.

Consumption under Uncertainty: Random Walk Hypothesis; Interest Rate and Saving; Consumption and Risky Asset: Consumption CAPM.

UNIT-III

(Investment Theories)

Investment and Investment Function: Type of Investment, Role of investment using Investment Multiplier, Classical and Keynesian Theories of Investment, Accelerator Theory of Investment, Neo-Classical Theory of Investment, and Tobin's-q Theory of Investment. Effects of Uncertainty, Kinked and Fixed Adjustment Costs, Investment in the Housing Market.

UNIT-IV

(Money Supply and Demand)

***Supply of Money*:** Theoretical Debate and Empirical Attempts to define money; Components of Supply of Money, Credit Creation by Commercial Banks, Money Multiplier.
***Demand for Money*:** Classical Quantity Theory, Keynesian Theory, Baumol and Tobin's Contributions. Friedman's Restatement of Quantity Theory of Money.

Recommended Readings:

Essential Readings:

Ackley, G. (2007). *Macroeconomic Theory*. Macmillan.

Allen, R.G.D. (1967). *Macro-Economic Theory: A Mathematical Treatment*. Palgrave Macmillan.

Blanchard, O. (2009). *Macroeconomics*. New Delhi: Pearson Education Inc.

Blanchard, O.J. & Fischer. (1996). *Lectures on Macroeconomics*, Delhi: PHI Learning Pvt. Ltd.

Branson, W. H. (1989). *Macro Economic Theory and Policy* (3rd ed.). Harper Collins.

D'Souza, E. (2008). *Macroeconomics.* Pearson Education.

Dornbusch, R., Fisher, S & Startz, R. (2002). *Macroeconomics*, (9th ed.). New Delhi: Tata McGraw Hills.

Levacic, R. & Rebmann, A. (1982). *Macroeconomics: An Introduction to Keynesian Neoclassical Controversies*. Macmillan.

Miller, R. L.& Pulsinelli, R. (1986). *Macroeconomics.* New York: Harper & Row.

Ola Olsson (2010). *Essentials of Advanced Macroeconomic Theory*, Routledge.

Romer, D. (1996). *Advanced Macroeconomics*. McGraw-Hill.

Further Readings:

Abel, A. B., Bernanke, B. S. (2007). *Macroeconomics.* New Delhi: Pearson Education Inc.

Dornbusch, R. (1980). *Open Economy Macroeconomics*. New York: Weltwirtschaftliches Archive.

Frisch, H. (1983). *Theories of Inflation*. Cambridge: Cambridge University Press.

Froyen, R. T. (2002). *Macro Economics.* New Delhi: Pearson Education Inc.

Gordon, R. J. (2012). *Macroeconomics*. Delhi: PHI Learning Pvt. Ltd.

Heijdra, B. J. (2017). *The Foundations of Modern Macroeconomics*. Oxford University Press.

Hicks, J. R. (1950). *A Contribution to the Theory of Trade Cycle* Oxford*: Clarendon Press*.

Lucas, R. (1981). *Studies in Business Cycle Theory*. , Massachusetts: MIT Press Cambridge.

Makin, A. J. (2002). *International Macroeconomics*. Princeton Hall.

Mankiw, N. G. & Romer, D. (1991). *New Keynesian Economics. MIT, 2 Volumes.*

Mathews, R. C. O. (1964). *The Trade Cycle.* Cambridge: Cambridge University Press.

Pentecost, E. (2000). *Macroeconomics – An Open Economy Approach.* MacMillan Press Ltd.

Raghbendra, J. (1964). *Contemporary Macroeconomics: Theory and Policy.* New Age International Publishers.

Sherffrin, S. (1999). *Rational Expectations.* Cambridge University Press.

Snowden, B & Vane, H. R. (2005). *Modern Macroeconomics: Its Origin, Development and Current Stat.,* UK: Edward Press.

Wendy, Carlin/Soskice, D. (2007). *Macroeconomics: Imperfections, Institutions and Policies*. Oxford University Press.

PAPER –MAECO–103: QUANTITATIVE METHODS –I

Max. Marks	: 100
Theory	: 80 Marks
Internal Assessment	: 20 Marks
Time	: 3 Hours
Teaching Hours	: 50

Objective:

The objective of this paper is to train the students in the use of mathematical and statistical tools in analyzing economic problems. The course content contains simple tools and techniques, thought necessary for data collection, presentation, analysis and drawing inferences about various statistical hypotheses.

The students would be evaluated at the end of each semester through subjective type questions/answers (both short and essay type). The scripts would be evaluated by the examiners having adequate postgraduate teaching experience in the paper/option concerned.

Pedagogy of the Course Work:

The course relies on a combination of lectures, solving problems, and discussion of numerical examples, academic articles or real-life situations. Teacher will assign topic for assignments on contemporary themes and issues from the syllabi. Special tutorials/contact hour for one-to-one student-teacher interactions.

INSTRUCTIONS FOR PAPER-SETTER AND CANDIDATES:

1. The syllabus of this paper has been divided into four units.
2. There shall be 9 questions in all.
3. The first question would be **compulsory,** shall be short answer type (word limit 25-30 each). It would carry 15 short questions, spread over the entire syllabus. The candidate will be required to attempt any 10 short answer type questions. Each short answer type question would carry 2 marks (10×2 = 20).
4. Rest of the paper shall contain four units. Each unit shall have two questions and the candidates shall be required to attempt one question from each unit – 4 in all. Each question shall carry 15 marks (15×4 = 60).
5. Candidates are allowed to use Simple, Non-Scientific and Non-Programmable Calculator in the Examination Hall.
6. It is compulsory for paper setter to mention about the requirement of log, anti log and statistical tables.

UNIT - I

Differentiation: Review of Simple differentiation; Partial and total derivatives; Homogenous function; Euler's theorem, Maxima and Minima of a function of two independent variable; Constrained optimization; Lagrange's method.

Application of Derivatives in Economics: Elasticity of demand; Partial elasticities; Competitive and Complementary Commodities; Optimization Problems in cost, Revenue etc.

UNIT – II

Growth Rate and its Measurement; Present Value and its Applications.

Matrices: Review of adjoint and inverse of matrices; Rank of Matrix; Linear Independence and Dependence of Vectors/Matrices; Solutions of Equations (upto three) by Matrix Inverse and Gauss elimination Methods; Consistency of Equations; Quadratic Forms – Types and Properties. Simple Economic Applications.

UNIT – III

Correlation and Regression Analysis: Multiple and Partial Correlation involving three Variables. The Linear Regression Model; Finding the Multiple Linear Regression Equation and Coefficients upto two explanatory variables: Interpretation of the Coefficients, R^2 and adjusted R^2.

Time Series Analysis: Definition and components. Methods of Trend Measurement: Quadratic, Exponential and Modified Exponential. Seasonal Indices by Methods of Simple Average, Ratio to moving Average and Ratio to Trend.

UNIT – IV

Index Numbers: Concepts of Price Relative, Quantity Relative and Value Relative. Laspeyer's, Paasche's and Fisher Index Numbers. Test of an Ideal Index Number. Base shifting, Splicing and deflating of Index Numbers. Problems in the Construction and Limitation of Index Numbers.

Probability Theory: Classical, empirical and Axiomatic Definitions of Probability. Laws of Addition and Multiplication (with examples); Bayes Theorem and its applications. Concept of Random Variable; probability mass and density functions; Expected Values, Moments (definition and types); Moments Generating Function (definition and properties).

Recommended Readings:

Essential Readings:

Allen, R. G. D. (latest edition). *Mathematica*l *Analysis for Economists*. New Delhi: Trinity Press.

Allen, R. G. D. (1986). *Mathematica*l *Analysis for Economists.* Macmillan Publisher India Ltd.

Arora, P. N., Arora, S., Arora, S. & Arora, A. (2007). *Comprehensive Statistical Methods.* New Delhi: S. Chand & and Company Ltd.

Chiang, A. C. & Wainwrigh, K. (2005). *Fundamental Methods of Mathematical Economics* (4th ed.). New York: McGraw Hill.

Devore, J. L. (2012). *Probability and Statistics for Engineering and the Sciences*, (8th ed.), Cengage Learning India.

Gupta, S. C. (latest edition). *Fundamentals of Statistics* (7th ed.). Himalaya Publications.

Spiegel, M. R. & Stephens L. J. (2000). *Theory and Problems of Statistics:* Schaum's Series, *(*3rd ed.). New Delhi: Tata McGraw Hill.

Weber, J. E. (1982). *Mathematical Analysis (Business and Economic Applications)* (4th ed.). New York: Harper and Row.

Yamane, T. (2006). *Mathematics for Economist-an Elementary survey* (2nd ed.). New Delhi: Prentice-Hall.

Yamane, T. (1972). *Fundamental Methods for Mathematical Economics*. New Delhi: Prentice-Hall.

Further Readings:

Freund, J. E. (1992). *Mathematical Statistics*. (5th ed.). New Delhi: Prentice-Hall, Cengage Learning, India.

Gupta, S. P. (2014). *Statistical Methods*. New Delhi: Sultan Chand & Sons.

Kandoi, B. (2012). *Mathematics for Business and Economics with Applications*, Vol. 1. Mumbai: Himalaya.

Simon, C. P. & Lawrence, B. (2006). *Mathematics for Economists* (Indian ed.). Viva –Norton Student edition: Norton and Company.

PAPER – MAECO-104: INTERNATIONAL ECONOMICS

Max. Marks	: 100
Theory	: 80 Marks
Internal Assessment	: 20 Marks
Time	: 3 Hours
Teaching Hours	: 50

Objective:

The course intends to provide a deep understanding about the broad principles and theories, which tend to govern the free flow of trade in goods, services and capital – both short-term and long-term – at the global level. The contents of the paper spread over different modules, lay stress on the theory and nature of the subject which, in turn, will greatly help them to examine the impact of the trade policies followed both at the national and international levels as also their welfare implications at macro level and the distribution of gains from trade.
The students would be evaluated at the end of each semester through subjective type questions/answers (both short and essay type). The scripts would be evaluated by the examiners having adequate postgraduate teaching experience in the paper/options concerned.

Pedagogy of the Course Work:
The course relies on a combination of lectures, solving problems, and discussing of academic articles or real life situations. Teacher will assign topic for assignments on contemporary themes and issues from the syllabi.

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES:

1. The syllabus of this paper has been divided into four units.

2. There shall be 9 questions in all.

3. The first question would be **compulsory**, shall be short answer type (word limit 25-30 each). It would carry 15 short questions, spread over the entire syllabus. The candidate will be required to attempt any 10 short answer type questions. Each short answer type question would carry 2 marks (10x2=20).
4. Rest of the paper shall contain 4 units. Each Unit shall have two questions and the candidates shall be required to attempt one question from each Unit – 4 in all. Each question shall carry **15** marks (15 x 4 = 60).

UNIT – I

***International Trade Theory*:**

Trade Based on Absolute Advantage: (Adam Smith), Comparative Advantage (David Ricardo) Advantage and Opportunity Costs (Haberler's theory, Gains from trade under constant cost as well as increasing costs).

Resources and Trade: Heckscher-Ohlin Model, Leontief Paradox.

Imperfect Competition and International Trade (Intra-industry trade), Trade Based on Dynamic Technological Differences (Technological Gap and Product Cycle Models)

UNIT – II

International Trade Policy*:*

Theory of Tariffs: Partial Equilibrium analysis of Tariff (both small country and large country case), General Equilibrium analysis of a Tariff (both small country and large country case). Optimum tariff.

Non-Tariff Barriers and Neo-protectionism.

Economic Integration: Theory of Customs Unions. Static effects (Trade creation and trade diversion). Dynamic effects of custom unions.

UNIT – III

***The Balance of Payments*:**

Concept and Components of Balance of Payment.

The Price Adjustment Mechanism with Flexible and Fixed Exchange Rates, Marshall-Learner conditions, J-curve effect, Gold Standard (Price-Species Flow Mechanism).

The Income Adjustment Mechanism, Foreign Trade Multiplier.

Open-Economy Macroeconomics and Adjustment Policies: Equilibrium in the Goods Market, in the Money Market and in the Balance of Payments (Mundell-Fleming Model).

UNIT – IV

Foreign Exchange Markets and International Monetary System:

Foreign Exchange Rates, Arbitrage, Spot and Forward Rates, Currency Swaps, Futures and Options, Foreign Exchange Risks, Hedging and Speculation. Euro currency Markets.

The International Monetary System: Past, Present and Future.

Recommended Readings:

Essential Readings:

Bhagwati, J. (1987). *International trade: Selected readings.* (*Vol. 1*). MIT Press.

Clement, M.O., Pfister, R. L., & Rothwell, K. J. (1967). *Theoretical issues in international economics*. Constolete and Company Ltd.

Dunn, R.M., & Mutti, J. H. (Latest Edition). *International economics*. New Jersey: John Wiley & Sons. Inc.

Jones, R. W., & Kenen, P.B. (1984). *Handbook of international economics. Vol. I, II and III.* New York: Elsevier.

Krugman, P.R., & Obstfeld, M.(Latest Edition). *International economics: Theory and policy*. United States: Pearson Publication.

Salvatore, D. (Latest Edition). *International economics*. New Jersey: John Wiley& Sons Inc.

Souderton, B., & Geoffrey R. (1994). *International economics*. London : McMillan.

Further Readings:

Ellsworth, P T & Clark L. (1975). *The International economy*. London: McMillan Press.

Mikic, M. (1998). *International trade*. London: McMillan Press.

Peter H L. (1969). *International economics.* New Jersey : Princeton University Press.

Shenkar, O., Luo, Y., & Chi, T. (2014). *International business*. London: Routledge.

SEMESTER–II

PAPER – MAECO-201: MICRO ECONOMICS-II

Max. Marks	: 100
Theory	: 80 Marks
Internal Assessment	: 20 Marks
Time	: 3 Hours
Teaching Hours	: 50

Objective:

The objective of this paper is to rigorously and comprehensively equip the students with theoretical concepts, methodology and process of reasoning involved in analyzing economic behavior of individuals, firms and markets using, in general, a static and partial equilibrium framework.

The students would be evaluated at the end of each semester through subjective type questions/answers (both short and essay type). The scripts would be evaluated by the examiners having adequate postgraduate teaching experience in the paper/option concerned.

Pedagogy of the Course Work :

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4. Rest of the paper shall contain 4 units. Each unit shall have two questions and the candidates shall be required to attempt one question from each Unit— 4 in all. Each question shall carry **15** marks ($15 \times 4 = 60$).

UNIT-I

Analysis of Competitive Markets: Profit Maximization and Equilibrium of a Firm and Industry in the Short Run and Long Run under Perfect Competition including the Supply Curve.
Consumers' and Producers' Surplus and Efficiency of Perfect Competition. Welfare Effects of an Excise Tax, Import Tariff, Price controls (supports) and Production Quotas.

Monopoly: Source, Measurement and Welfare Costs and Dynamic benefits of Monopoly Power. Monopoly Price Discrimination. Output and Welfare Comparisons with Single Price Monopoly. Inter–temporal Price Discrimination and Peak Load Pricing. Two part tariffs and Tie-in Sales.
Public Policy toward Monopoly

UNIT-II

Monopolistic Competition: Characteristics. A firm's Short and Long Run Equilibrium with Differentiated Products and Selling Costs (Advertisement). Group equilibrium. The Issue of Excess Capacity under Monopolistic Competition. The debate over Advertising.

Oligopoly: Cournot, Stackellberg and Kinked demand Curve. Leadership Models. Collusion and Cartels.
Game Theory and Economics of Cooperation: Dominant Strategy and Nash Equilibrium. The Prisoner's Dilemma, Mixed Strategies, Repeated Games, Collusion and Tit-for-Tat Strategy. Public Policy towards Oligopoly. Contestable Markets.

UNIT-III

Markets for Factor Inputs:

Factor Pricing under Perfect Competition:
Demand for a Factor Input when One Input is Variable. Demand for a Factor Input when Several Inputs are Variable. Market Demand for a Labour. Supply of Labour by an Individual: Trade Off between Work and Leisure. Wage rate determination.

Factor Pricing under Imperfect Competition:
Wage rate determination with Monopolistic Power in Product Market, Monoposonistic Power in Factor Market; Bilateral Monopoly.
Competitive Buyer Firm versus Monopoly Union.

UNIT-IV

Welfare Economics:
Social Welfare Function: Utilitarian, Rawlsian and Bergson-Samuelson Social Welfare Functions and Optimization of Social Welfare.
General Equilibrium and Efficiency: Pareto-Optimality Conditions. Perfect Competition and Economic Efficiency. First and Second Theorem of Welfare Economics. Compensation Criteria and Arrow's Impossibility Theorem.
Externalities & Efficiency: Externalities in Production and Consumption. Coase Theorem and Public Policies towards Externalities. Common Property: Overuse of Common Property, Solving Common Problems.

Recommended Readings:

Essential Readings:

Boumol, W. J. & Blinder, A. S. (2007*). Macroeconomics: Principles and Policy,* (9th ed.). Thomson.

Ferguson, C. E. and Gold J. P. (1989). *Micro Economic Theory* (6th ed.). All India Book Sellers.

Mankiw, G. N. (2017). *Principles of Economics,* (3rd ed.). Thomson.

Pindyck, Robert, S., Rubinfel, D. L. & Gupta, P.L. (2017). *Micro-Economics.* New Delhi: Prentice Hall of India.

Salvatore, D. (2007). *Microeconomics: Theory and Applications* (4th ed.). Oxford University Press.

Further Readings:-

Briet, W. & Hochman, H.M. (1977). *Readings in Micro Economics*. Hindsala: Dryden Press.

Koutsoyiannis, A. (2016*). Modern Micro Economics*. London: McMillan Press.

Maddala, G. S. & Miler, E. (2004*). Micro-Economic Theory and Applications*. Tata McGraw Hill.

Ryan, W. J. L. & Pearce, D. W. (1989). *Price Theory*. New Delhi : McMillan India.

PAPER – MAECO-202: MACRO ECONOMICS-II

Max. Marks	: 100
Theory	: 80 Marks
Internal Assessment	: 20 Marks
Time	: 3 Hours
Teaching Hours	: 50

Objective:

Macroeconomics or aggregative economics analysis establishes the functional relationship between the large aggregates. The aggregate analysis has assumed such a great significance in recent times that a prior understanding of macroeconomic theoretical structure is considered essential for the proper comprehension of the different issues and policies. Macroeconomics now is not only a scientific method of analysis; but also a body of empirical economic knowledge.

The paper entitled "Macro-Economics-I" equips the students at the postgraduate level to understand systemic facts and latest theoretical developments for empirical analysis.

The students would be evaluated at the end of each semester through subjective type questions/answers (both short and essay type). The scripts would be evaluated by the examiners having adequate postgraduate teaching experience in the paper/option concerned.

Pedagogy of the Course Work :

The course relies on a combination of lectures, solving problems, and discussing of academic articles or real-life situations. Teacher will assign topic for assignments on contemporary themes and issues from the syllabi. Special tutorials/contact hour for one-to-one student-teacher interactions.

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES:

1. The syllabus of this paper has been divided into four units.
2. There shall be 9 questions in all.
3. The first question, which would be **compulsory**, shall be short answer type (word limit 25-30 each). It would carry 15 short questions, spread over the entire syllabus. The candidate will be required to attempt any 10 short answer type questions. Each short answer type question would carry 2 marks ($10 \times 2 = 20$).
4. Rest of the paper shall contain 4 units. Each unit shall have two questions and the candidates shall be required to attempt one question from each Unit — 4 in all. Each question shall carry **15** marks ($15 \times 4 = 60$).

UNIT-I

(Simultaneous Equilibrium & Tatonnement Process under Closed and Open Economies)

IS and LM Framework: Derivation, Properties, Shifts, and Rotations of IS and LM Curves under closed and open economy systems. Derivation, Properties, Shifts, and Rotations of BP Curve. Simultaneous Equilibrium in Money and Product Markets. Impact of Opening-up on simultaneous equilibrium (i.e., IS-LM-BP simultaneous equilibrium).

Monetary and Fiscal Policies: Objectives, Conflicts among Objectives. Relative Effectiveness of Monetary and Fiscal Policies under Different Situations in IS-LM-BP Framework. International Trinity and Quadrilemma choices under IS-LM-BP framework.

UNIT-II

(Inflation Theories and Philips Curve)

Inflation: Effects of Inflation. Theories of Inflation: Quantity Theory, Keynesian Theory, Monetarist views on Inflation, Modern theory of Inflation, Structural Theory. The menu of policy choices: Philips Curve Analysis –Short Run and Long Run views. The Monetarist-Keynesian Debate and the Phillips Curve.

UNIT-III

(Trade Cycle Models/Theories)

Trade Cycle: Hansen–Samuelson Accelerator–Multiplier Interaction Model, Hicks Model, Kaldor Model, Goodwin model of endogenous cycles.

UNIT – IV

(The New Macroeconomics)

The New Classical School: Rational Expectations Hypothesis: Dynamic Time Inconsistency, Policy Ineffectiveness Proposition.
The Random Walk of GDP: The Relative Importance of AD and AS.
Real Business Cycle Model: Disturbances and Propagation mechanism. Macroeconomic Policy in Real Business Cycle Model.
The New-Keynesian School: Real and Nominal Wage-Price Rigidity Models - Menu Costs Model, Implicit Wage Contract Models, Efficiency Wage Models, Insider-Outsider Models.

Recommended Readings:

Essential Readings:

Ackley, G. (2007). *Macroeconomic Theory*. Macmillan.

Allen, R.G.D. (1967). *Macro-Economic Theory: A Mathematical Treatment*. Palgrave Macmillan.

Blanchard, O. (2009). *Macroeconomics*. New Delhi: Pearson Education Inc.

Blanchard, O.J. & Fischer. (1996). *Lectures on Macroeconomics*, Delhi: PHI Learning Pvt. Ltd.

Branson, W. H. (1989). *Macro Economic Theory and Policy* (3rd ed.). Harper Collins.

D'Souza, E. (2008). *Macroeconomics.* Pearson Education.

Dornbusch, R., Fisher, S & Startz, R. (2002). *Macroeconomics*, (9th ed.). New Delhi: Tata McGraw Hills.

Levacic, R. & Rebmann, A. (1982). *Macroeconomics: An Introduction to Keynesian Neoclassical Controversies*. Macmillan.

Miller, R. L.& Pulsinelli, R. (1986). *Macroeconomics.* New York: Harper & Row.

Ola Olsson (2010). *Essentials of Advanced Macroeconomic Theory*, Routledge.

Romer, D. (1996). *Advanced Macroeconomics*. McGraw-Hill.

Further Readings:

Abel, A. B., Bernanke, B. S. (2007). *Macroeconomics.* New Delhi: Pearson Education Inc.

Dornbusch, R. (1980). *Open Economy Macroeconomics*. New York: Weltwirtschaftliches Archive.

Frisch, H. (1983). *Theories of Inflation*. Cambridge: Cambridge University Press.

Froyen, R. T. (2002). *Macro Economics.* New Delhi: Pearson Education Inc.

Gordon, R. J. (2012). *Macroeconomics*. Delhi: PHI Learning Pvt. Ltd.

Heijdra, B. J. (2017). *The Foundations of Modern Macroeconomics*. Oxford University Press.

Hicks, J. R. (1950). *A Contribution to the Theory of Trade Cycle* Oxford*: Clarendon Press*.

Lucas, R. (1981). *Studies in Business Cycle Theory*. , Massachusetts: MIT Press Cambridge.

Makin, A. J. (2002). *International Macroeconomics*. Princeton Hall.

Mankiw, N. G. & Romer, D. (1991). *New Keynesian Economics. MIT, 2 Volumes.*

Mathews, R. C. O. (1964). *The Trade Cycle*. Cambridge: Cambridge University Press.

Pentecost, E. (2000). *Macroeconomics – An Open Economy Approach.* MacMillan Press Ltd.

Raghbendra, J. (1964). *Contemporary Macroeconomics: Theory and Policy.* New Age International Publishers.

Sherffrin, S. (1999). *Rational Expectations.* Cambridge University Press.

Snowden, B & Vane, H. R. (2005). *Modern Macroeconomics: Its Origin, Development and Current Stat.,* UK: Edward Press.

Wendy, Carlin/Soskice, D. (2007). *Macroeconomics: Imperfections, Institutions and Policies*. Oxford University Press.

PAPER – MAECO-203: QUANTITATIVE METHODS – II

Max. Marks	: 100
Theory	: 80 Marks
Internal Assessment	: 20 Marks
Time	: 3 Hours
Teaching Hours	: 50

Objective:

The objective of the paper is to rigorously and comprehensively equip the students with theoretical concepts, methodology and process of reasoning involved in analyzing economic behavior of individuals, firms and markets using, in general, a static and partial equilibrium framework.

The students would be evaluated at the end of each semester through subjective type questions/answers (both short and essay type). The scripts would be evaluated by the examiners having adequate postgraduate teaching experience in the paper/option concerned.

Pedagogy of the Course Work:

The course relies on a combination of lectures, solving problems, and discussion of numerical examples, academic articles or real-life situations. Teacher will assign topic for assignments on contemporary themes and issues from the syllabi. Special tutorials/contact hour for one-to-one student-teacher interactions.

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES:

1. The syllabus of this paper has been divided into four units.
2. There shall be 9 questions in all.
3. The first question would be **compulsory**, shall be short answer type (word limit 25-30 each). It would carry 15 short questions, spread over the entire syllabus. The candidate will be required to attempt any 10 short answer type questions. Each short answer type question would carry 2 marks (10x2=20).
4. Rest of the paper shall contain 4 units. Each Unit shall have two questions and the candidates shall be required to attempt one question from each Unit – 4 in all. Each question shall carry **15** marks (15 x 4 = 60).
5. Candidates are allowed to use Simple, Non-Scientific and Non-Programmable Calculator in the Examination Hall.
6. It is compulsory for paper setter to mention about the requirement of log, anti log and statistical tables.

UNIT – I

Integration: Indefinite and Definite, Integration of a Function of only one Variable; Methods of Integration; Substitution and Partial Fraction. Application of Integration in Economics; Consumers' and Producers' Surplus; Finding Demand Function, Maximum Profit and Capital Growth Equation.

Differential Equations: Introduction, Solution of Differential equations – Variable-separable, Homogenous, Exact, Linear and Bernoulli's form cases; Solution of Linear Differential Equation with Constant Coefficients.

Economic Applications w.r.t. Demand, Revenue, Cost Functions and Simple Market Equilibrium Models.

UNIT – II

Difference Equations: Solution of a Difference Equation and Simple Economic Applications.

Linear Programming: Definition, Assumptions, Formulation and Solution of LPP (Feasible, Basic and Optimal Solutions) by graphical and simplex method. Dual Problem: Solution of Primal and Dual Problems by Simplex Method and their Interpretation.

UNIT – III

Probability Distributions: Properties (including derivation of mean and variance) and Applications of Binominal, Poisson and Normal Distributions.

Statistical Inferences: Concepts of parameter, statistic and standard error (including its utility). Sampling Distribution of a Statistic, Distribution of Sample Mean, Properties of an Ideal Estimator; Unbiasedness, Consistency, Efficiency and Minimum Variance. Interval Estimation and Confidence Interval for mean, difference of means, single proportion and difference of proportions.

UNIT – IV

Tests of Significance: Large and Small Sample Tests based on Normal and t Distribution, Test of Significance of Mean, Difference between means, Single Proportion, Difference between two proportions, Simple Correlation Coefficient and Simple Regression Coefficient, test based on F and Z Distribution (Fisher's Z Statistics).

Chi-Square Tests: Conditions for Chi-Square Test; Application of χ^2 Distribution – Test for Goodness of Fit, Independence and Specified Value of the Variance.

Analysis of Variance: Applications with regard to one way and two way classifications.

Recommended Readings:

Essential Readings:

Allen, R. G. D. (1986). *Mathematica*l *Analysis for Economists.* Macmillan Publisher India Ltd.

Allen, R. G. D. (latest edition). *Mathematica*l *Analysis for Economists*. New Delhi: Trinity Press.

Arora, P. N., Arora, S., Arora, S. & Arora, A. (2007). *Comprehensive Statistical Methods.* New Delhi: S. Chand & and Company Ltd.

Chiang, A. C. & Wainwrigh, K. (2005). *Fundamental Methods of Mathematical Economics* (4th ed.). New York: McGraw Hill.

Devore, J. L. (2012). *Probability and Statistics for Engineering and the Sciences*, (8th ed.), Cengage Learning India.

Gupta, S. C. (latest edition). *Fundamentals of Statistics* (7th ed.). Himalaya Publications.

Spiegel, M. R. & Stephens L. J. (2000). *Theory and Problems of Statistics:* Schaum's Series, *(*3rd ed.). New Delhi: Tata McGraw Hill.

Weber, J. E. (1982). *Mathematical Analysis (Business and Economic Applications)* (4th ed.). New York: Harper and Row.

Yamane, T. (1972). *Fundamental Methods for Mathematical Economics*. New Delhi: Prentice-Hall.

Yamane, T. (2006). *Mathematics for Economist-an Elementary survey* (2nd ed.). New Delhi: Prentice-Hall.

Further Readings:

Freund, J. E. (1992). *Mathematical Statistics.* (5th ed.). New Delhi: Prentice-Hall, Cengage Learning, India.

Gupta, S. P. (2014). *Statistical Methods.* New Delhi: Sultan Chand & Sons.

Kandoi, B. (2012). *Mathematics for Business and Economics with Applications*, Vol. 1. Mumbai: Himalaya.

Simon, C. P. & Lawrence, B. (2006). *Mathematics for Economists* (Indian ed.). Viva –Norton Student edition: Norton and Company.

PAPER – MAECO-204: PUBLIC FINANCE

Max. Marks	: 100
Theory	: 80 Marks
Internal Assessment	: 20 Marks
Time	: 3 Hours
Teaching Hours	: 50

Objectives:

Role and functions of the Government in an economy have been changing with the passage of time. The term 'Public Finance' has traditionally been applied to the package of those policies and operations which involve the use of tax and expenditure measures while budgetary policy is an important part to understand the basic problems of use of resources, distribution of income etc. There are vast array of fiscal institutions – tax systems, expenditure programmes, budgetary procedures, stabilization instruments, debt issues, levels of government etc., which raise a spectrum of issues arising from the operation of these institutions. Further, the existing of externalities, concern for adjustment in the distribution of income and wealth, etc. require political processes for their solution in a manner which combines individual freedom and justice. This paper combines a thorough understanding of fiscal institutions with a careful analysis of the issues which under line budgetary policies in general and Indian experience in particular.

The students would be evaluated at the end of each semester through subjective type questions/answers (both short and essay type). The scripts would be evaluated by the examiners having adequate postgraduate teaching experience in the paper/options concerned.

Pedagogy of the Course Work:
The course relies on a combination of lectures, solving problems, and discussing of academic articles or real life situations. Teacher will assign topic for assignments on contemporary themes and issues from the syllabi.

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4. Rest of the paper shall contain 4 units. Each Unit shall have two questions and the candidates shall be required to attempt one question from each Unit – 4 in all. Each question shall carry **15** marks (15 x 4 = 60).

UNIT – I

Comparison of Provision of Private Goods and Public Goods in General Equilibrium (Pareto's Optimality criteria.)
Equity in Distribution. Various approaches to distributive Justice.
Public Choice and Fiscal Policies. Voting rules.

UNIT – II

Various Approaches of Equity in Taxation: Benefit Principle including Lindahl Theory. Ability to Pay Approach.
Incidence analysis of taxation in various markets.
Effects of Taxation on Work Effort, Savings and Investment.

UNIT - III

Deficit Financing: Concept and its relation with Inflation, Deficit Financing in India.
Issues relating to Public Debt: Debt Burden Analysis and Management of Public Debt, Domar's concept of Debt Sustainability Public Debt in India.
Need for rule based fiscal consolidation. Fiscal Responsibility and Budget management (FRBM) act, 2003. Recent amendments to FRBM act.

UNIT – IV

Theories of Public Expenditure: Wagner's Law and Peacock - Wiseman Hypothesis. Structure and Classification of Public expenditure in India.
Principles of Multiunit Finance (Central. State and regional level)
Centre – State Financial Relations in India: Assessment of Horizontal and vertical imbalances. Role of Finance commissions.

Essential Readings:

Atkinson, A.B., & Stiglitz, J.E.(2007). *Lectures on public economics*. New Jersey: Princeton University Press.

Bhatia, H. L. (2006). *Public finance*. Vikas Publishing House Private Ltd.

Cullis, J., Jones, P., & Jones, P.R. (2009). *Public finance and public choice: Analytical perspectives*. New York: Oxford University Press.

Musgrave, R.A., & Musgrave, P.B. (1989). *Public finance in theory and practice*. New York: McGraw Hill Book Company.

Rao, M.G., & Sen, T. K. (1996). *Fiscal federalism in India: Theory and practice*. Macmillan India.

Recommended Readings:

Aronson, J.R. (1985). *Public finance*. New York: McGraw-Hill International.

Basu, K., & Maertens, A. (2012). *The new Oxford companion to economics in India*. New York: Oxford University Press.

Brown, C.V., & Jackson, P. M. (1990). *Public sector economics*. United Kingdom: Wiley-Blackwell.

Buchanan, J. M. (1958). *Public principles of public debt*. Homewood, Illinois: Richard D. Irwin.

Buchanan, J. M., & Musgrave, R. A. (1999). *Public finance and public choice: Two contrasting visions of the state*. Cambridge, Massachusetts: MIT Press.

Chelliah, R.J. (1996). *Towards sustainable growth: Essays in fiscal and financial sector reforms in India*. New York: Oxford University Press.

Datt, R., & Sundaram, K.P.M. (2003). *Indian economy*. New Delhi: S. Chand & Company.

Houghton, R. W. (1970). *Public finance: Selected readings*. Penguin.

Ulbrich, H.H. (2013). *Public finance in theory and practice Second edition*. London: Routledge.

Kurian, N. J. (1999). State government finances: A survey of recent trends. *Economic and Political Weekly*, *34* (19), 1115-1134.

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